

Indian Institute of Technology Hyderabad
Signal Processing, Fall 2026
Mid-term Examination, 16.09.2026, 6:00 pm to 8:00 pm, 20 points

“The important thing is not to stop questioning. Curiosity has its own reason for existing.” – *Albert Einstein*

Table 1: “Cheat Table:” Note that the DTFT and z-transform summations are subject to convergence conditions. In the definition of systems, $y[n] = T\{x[n]\}$ is the output of the system for input $x[n]$. For LTI systems, $h[n] = T\{\delta[n]\}$.

Signals	$\delta[n] = \begin{cases} 1, n = 0 \\ 0, n \neq 0 \end{cases}$	$u[n] = \begin{cases} 1, n \geq 0 \\ 0, n < 0 \end{cases}$	$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$
Analysis	$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$	$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$	$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}; z = re^{j\omega}$
Synthesis	$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}nk}$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega$	Not discussed
System	Memoryless	Linear	Time-invariant
Definition	$T\{x[n]\}$ depends only on $x[n]$	$T\{\alpha x_1[n] + \beta x_2[n]\} = \alpha T\{x_1[n]\} + \beta T\{x_2[n]\}$	$T\{x[n-n_0]\} = y[n-n_0]$
System	Causal	BIBO Stable	Linear Time-invariant
Definition	$T\{x[n]\}$ depends only on present and past inputs	$ x[n] < B_x \implies y[n] < B_y$	Linear and Time-invariant
LTI Convolution	$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$	$h_{lp}[n] = \frac{1}{2}(\delta[n] + \delta[n-1])$	$h_{hp}[n] = \frac{1}{2}(\delta[n] - \delta[n-1])$

1. Let's warm up.

- Given an example of a bounded signal for which the DTFT doesn't exist. (1)
- The z-transform is defined for points in the complex plane where the summation converges. This is called the region of convergence (ROC). What can you say about the shape of the ROC? (1)
- From first principles, derive the convolution sum for LTI systems. (1)

2. Onto signal denoising. How will you denoise the following 1-D signals with the specified noise spectrum using LTI systems? Justify.

- A signal corrupted with noise at $\omega = 0$. (1)
- A signal corrupted with noise at $\omega = \pi/2$. (2)
- A signal corrupted with noise at $\omega \geq \pi/2$. (2)
- A signal corrupted with noise at $\omega \leq \pi/2$. (2)

3. Fun in the frequency domain.

- Can a non-causal LTI system have linear phase? Explain. (2)
- Let $y[n] = x[2n]$, be a compressor system. Express the DTFT of $y[n]$ in terms of the DTFT of $x[n]$. Hint: express the DTFT in terms of the odd and even indices and see how you can eliminate the odd indices. (3)

4. $h_{lp}[n]$ and $h_{hp}[n]$ are more than meet the eye. Let's explore.

- $x[n] = [a, b, c, d]$, with $x[0] = a, x[1] = b, x[2] = c, x[3] = d$. Find $y_{lp}[n] = x[n] * h_{lp}[n]$, $y_{hp}[n] = x[n] * h_{hp}[n]$, where $*$ is the convolution operator. Zero pad the signal at the boundaries. (1)
- Consider the system shown below with two parallel branches. The top branch is composed of $h_{lp}[n]$ followed by a compressor, upsampler and $h_1[n]$. The bottom branch is composed of $h_{hp}[n]$ followed by a compressor, upsampler and $h_2[n]$. The compressor (denoted by $\downarrow 2$) is the system $y[n] = x[2n]$. The upsampler (denoted by $\uparrow 2$) is defined as follows: $y[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$.

Can you find $h_1[n]$ and $h_2[n]$ such that $x[n]$ can be recovered exactly (but perhaps with a delay)? Show your work. Use $x[n]$ from the previous question as a test signal. (4)

